

GeoLabeller

How-To

Software version v2.4.1

Generated 2026-10-02

A guide to labelling imagery with GeoLabeller, in two parts: placing and managing labels on the map, and recording what each label shows in the Snippet Editor. Every picture in it was taken from the running software by the script that builds this document, so the pictures show the version named above.

The imagery in the pictures is synthetic: flat 'water' with a bright rectangle for each vessel and a dark one for its shadow. Your imagery will look different; the controls will not.

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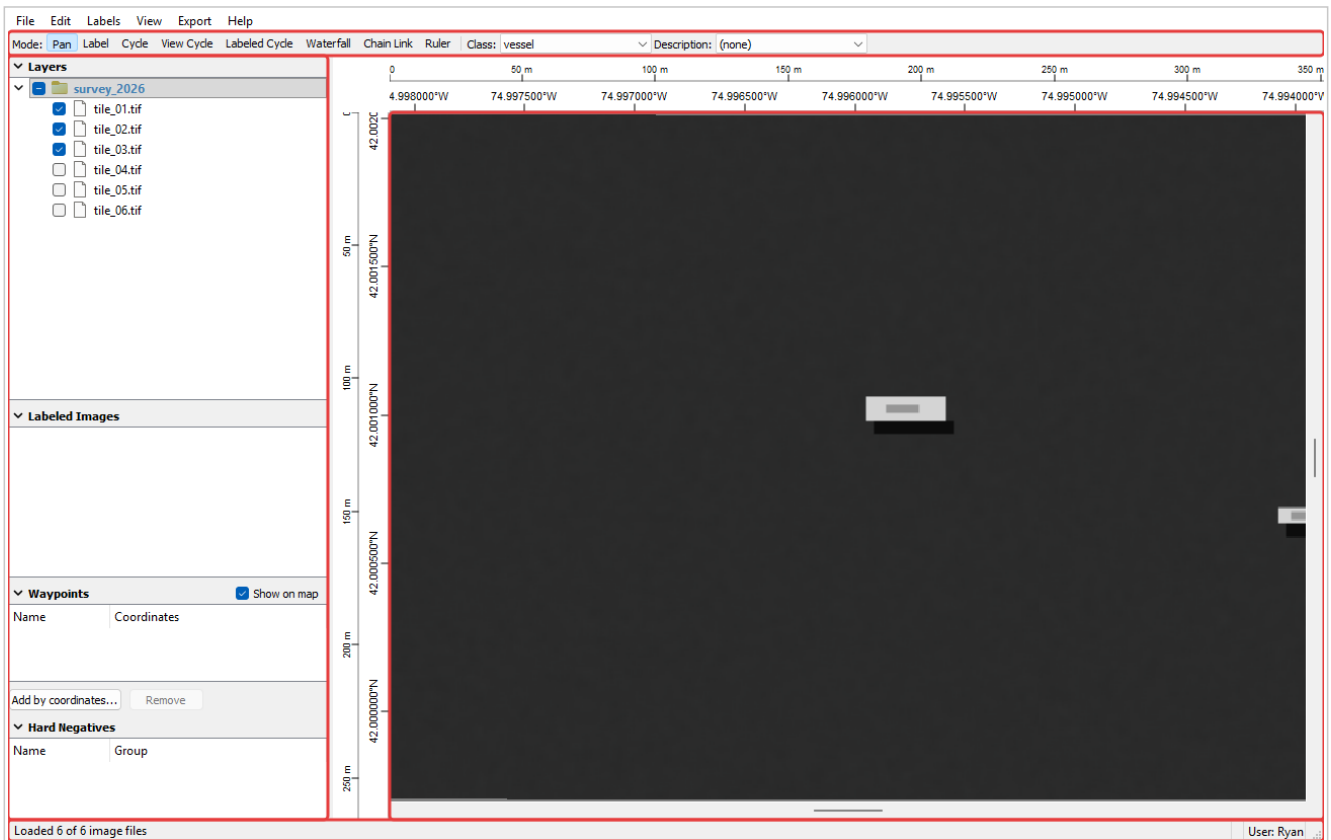
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Part 1

Labelling imagery on the map

1 The window

GeoLabeller opens with an empty project. Everything you do is recorded in the project, which you save to a file (File > Save Project); the imagery itself is never copied, only referred to.

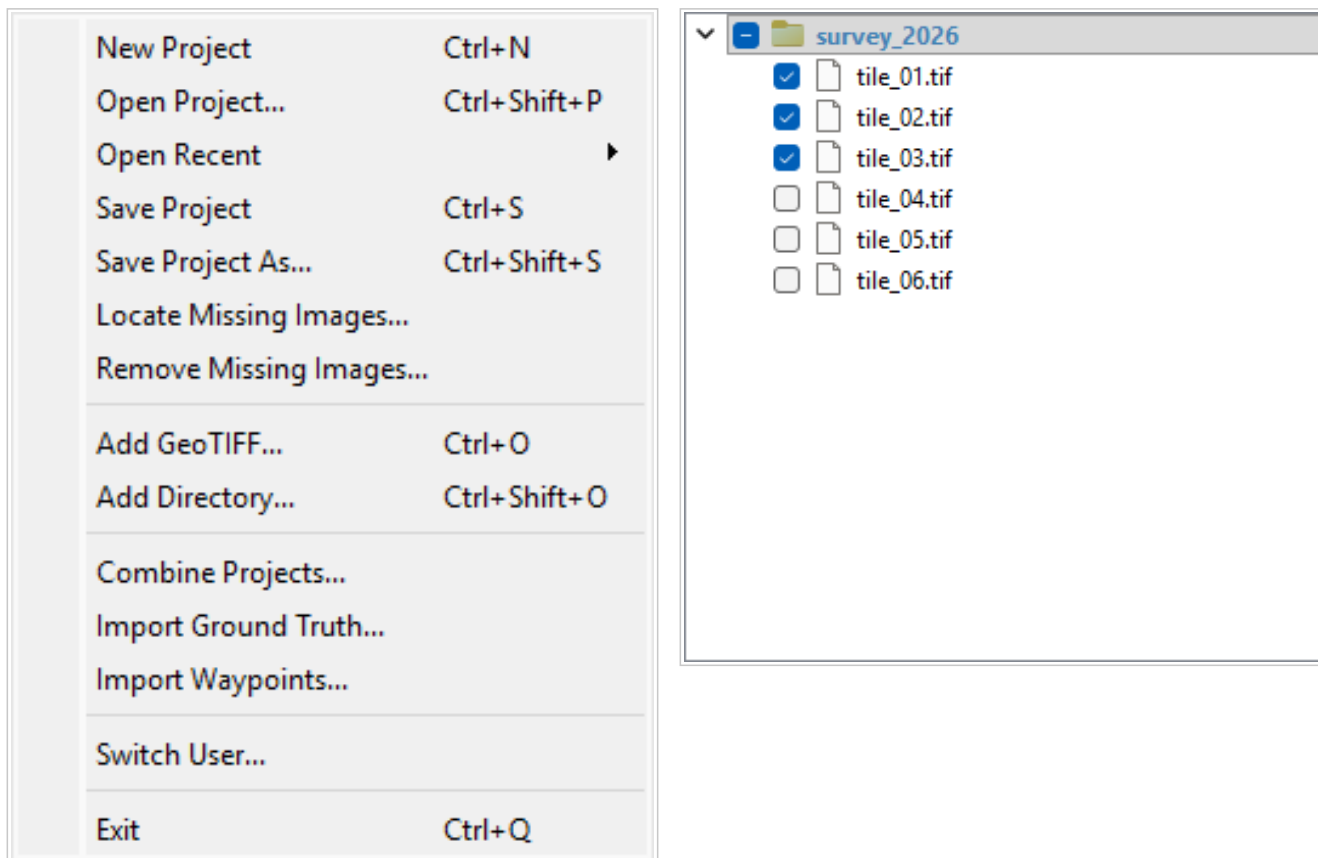


The main window with a folder of imagery loaded. Left: the panels. Top: the mode toolbar. Middle: the map. Bottom: the status bar, which says what the current mode wants from you.

- The LEFT COLUMN holds four panels that fold under their headers: Layers (the imagery, by folder), Labeled Images (every image with a label on it), Waypoints, and Hard Negatives.
- The TOOLBAR chooses the mode - what a click on the map does - and the class and description preset given to new labels.
- The MAP shows the imagery that is switched on, georeferenced imagery in its place on the ground. Drag to pan, wheel to zoom.
- The STATUS BAR answers 'what happens if I click now?' and reports what the last action did. Glance at it when unsure.
- The user name at the right of the status bar is recorded on every label, mask, rating and review you make (File > Switch User... to change it).

2 Loading imagery

1. File > Add Directory... and choose the folder. Every GeoTIFF in it (and in its sub-folders) is loaded, as a group named after the folder. File > Add GeoTIFF... loads single files.
2. The folder appears in the Layers panel. Tick an image to show it; tick the group to show them all. Nothing is shown until it is ticked, so a large survey does not fill the screen at once.
3. Right-click an image > Zoom to layer puts it in front of you.



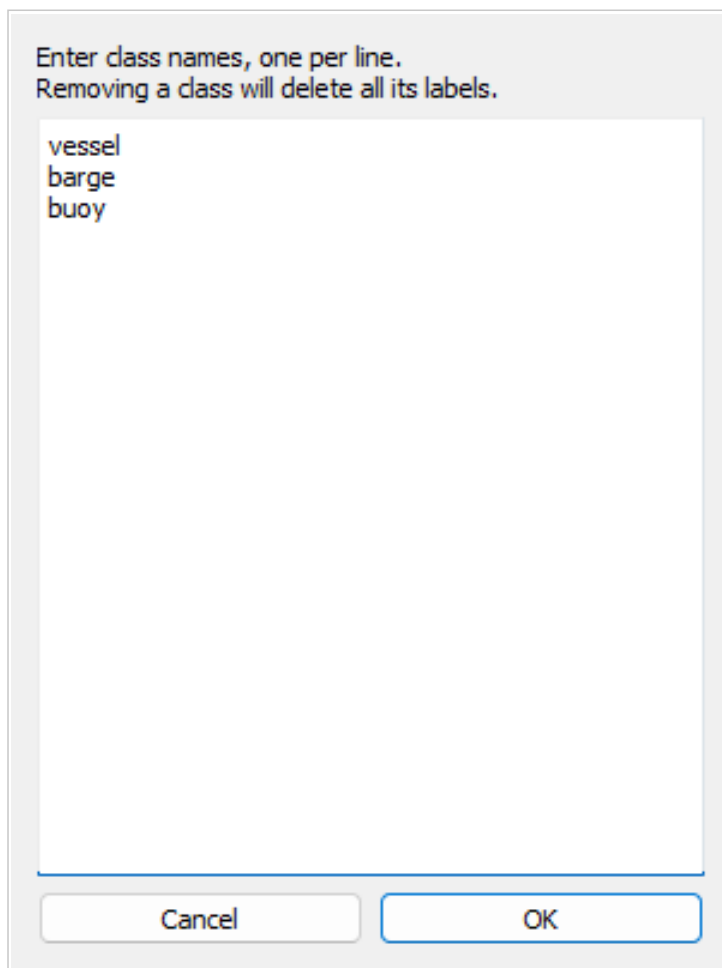
The File menu, and the Layers panel with a folder loaded and three of its images ticked.

Imagery without georeferencing is loaded too, under a 'Non-Georeferenced' root, and laid out side by side rather than on the ground. Labels on it have pixel positions but no latitude and longitude.

How a group is drawn - which bands, brightness, contrast and gamma - is set per group under View > Display Settings... (Ctrl+D), or by right-clicking the group. It changes the display only; nothing you record is affected. See section 8.

3 Classes

A label has a class: what the thing is. Set the classes up once under Labels > Edit Classes..., then pick the class for the next labels from the Class box on the toolbar - or press its number: 1 for the first class in the list, 2 for the second, and so on.

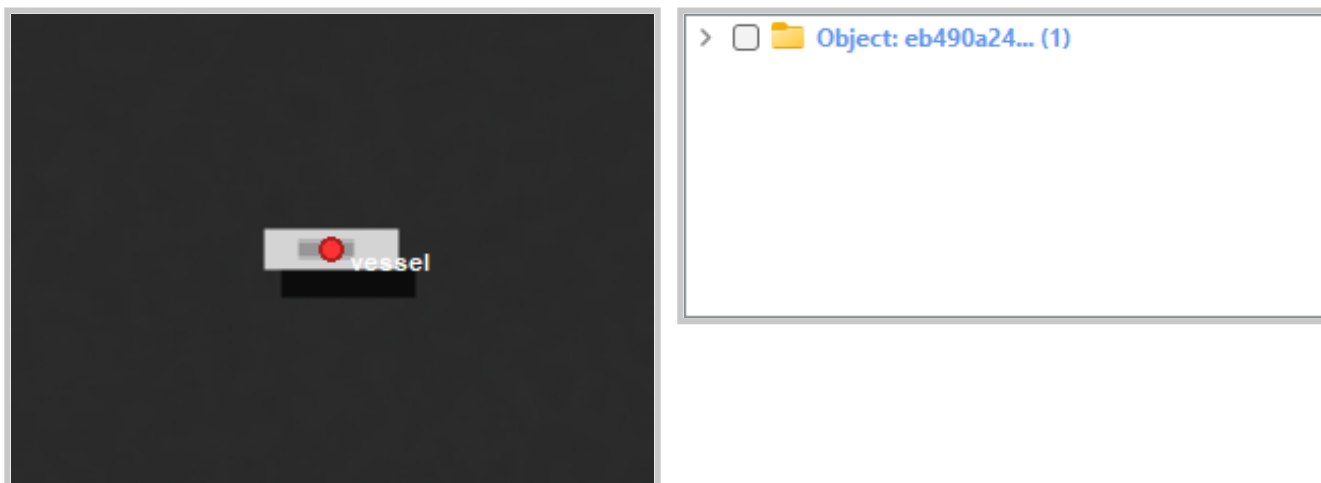


Labels > Edit Classes...

Removing a class removes its labels; the dialog says how many before it does. The change is one undo step.

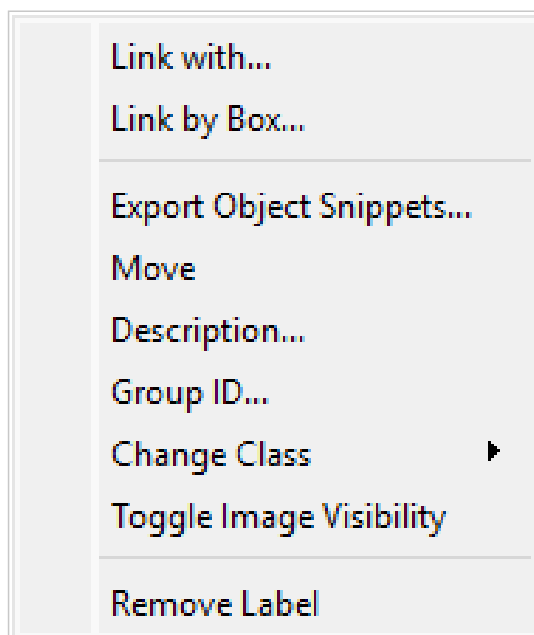
4 Placing labels

1. Press L (or click Label on the toolbar). The status bar now says a click places a label.
2. Click the object. A marker appears in the class's colour, and the image joins the Labeled Images panel.
3. Keep clicking objects. Space and Ctrl+Space step through the images in a cycle mode (section 6), so a pass over a survey is click, Space, click, Space.



A label just placed, and the Labeled Images panel listing it. The panel's checkboxes show and hide images too.

Right-click a label for everything that can be done to it: link it to the same object in another image, move it, describe it, name its group, change its class, or remove it.



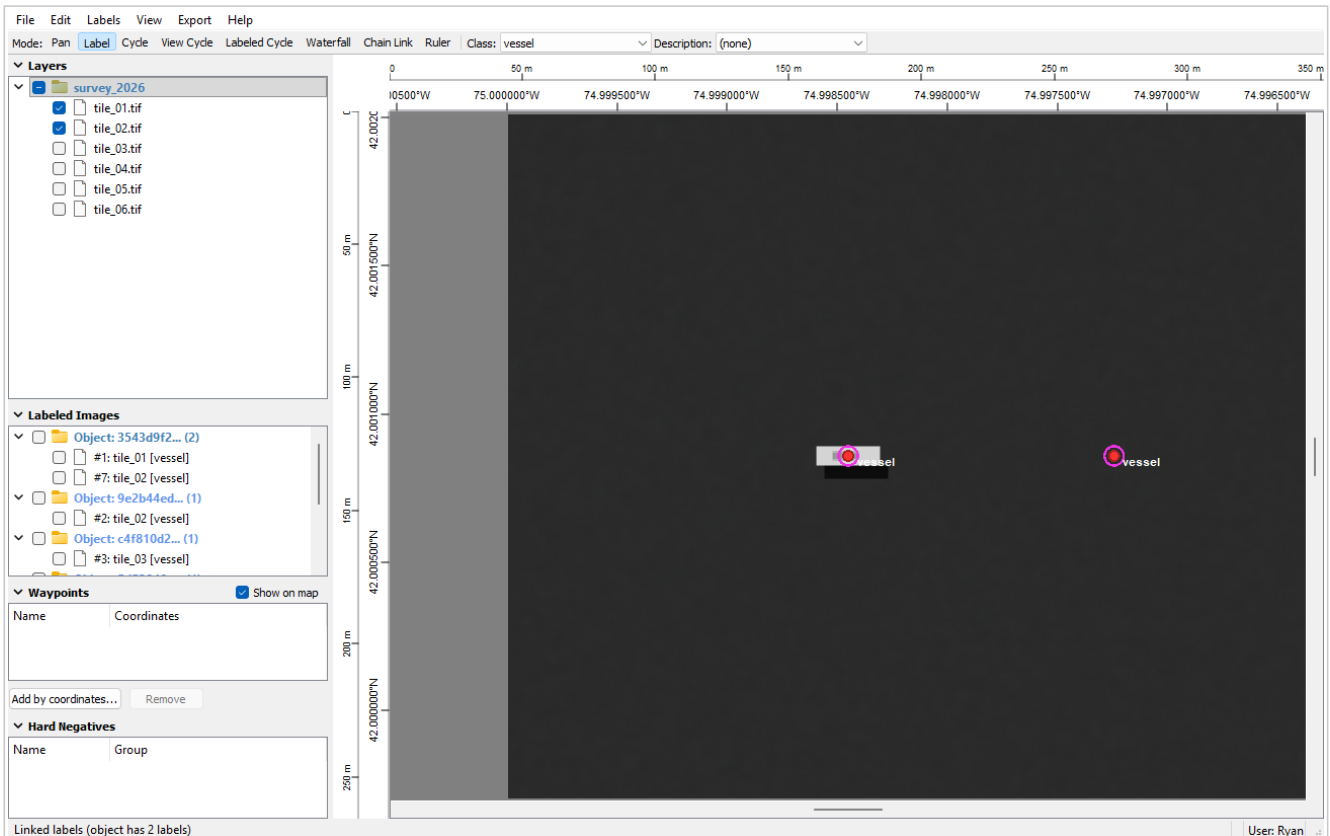
A label's right-click menu.

The Description box on the toolbar holds preset descriptions (Labels > Edit Descriptions...). Whichever is chosen is given to every label you place - Shift+1 to Shift+9 switch presets, Shift+0 chooses none. A description can also be typed per label, from its menu or in the Snippet Editor.

5 Linking labels: one object in several images

Where surveys overlap, the same vessel appears in more than one image. Link those labels and GeoLabeller knows they are one object: it is exported once per view but counted once, a heading or size measured on one view can be given to the others, and the Snippet Editor shows every view of the object together.

1. Right-click a label > Link with..., then click the other label. Or right-click > Link by Box... and drag a box round all the labels of one object.
2. For many objects in a row, press K (Chain Link): click the labels of one object, press N to start the next, Escape to finish.
3. Linked labels wear a halo in a shared colour. Right-click > Unlink takes one out of its object.



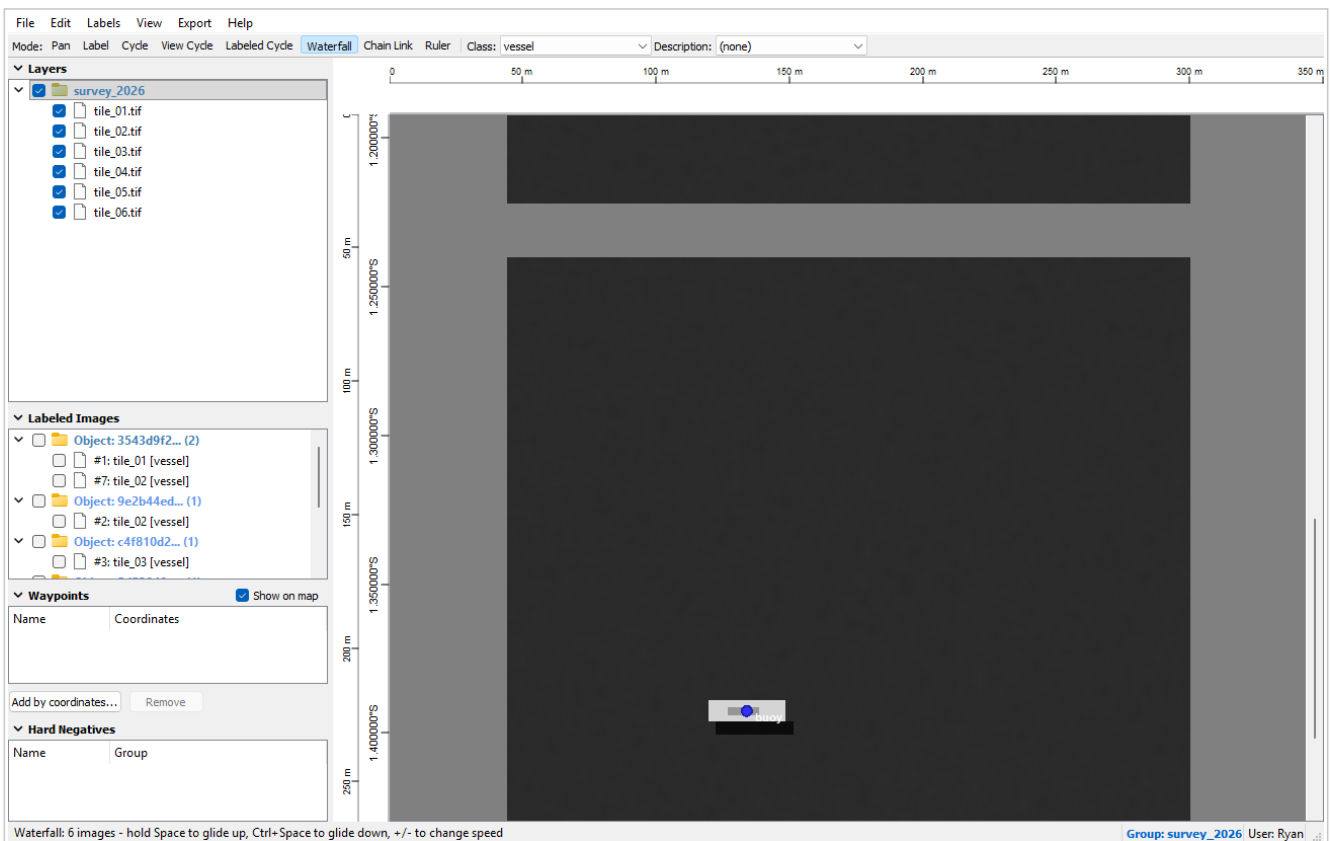
Two overlapping images with a linked pair: the same vessel labelled in each, haloed in one colour. Labels on the image in front are bracketed.

A Group ID (right-click > Group ID...) is a name for the whole linked object - a hull number, say - and is given to every label linked to it.

6 Getting around

The cycle modes step through a group's images one at a time, showing only the current one, so a survey of hundreds of tiles can be worked through in order. Select the group in the Layers panel first. In every cycle mode a left click still places a label, and the status bar says where you are.

Key	Does
C	Cycle: step through every image in the selected group. Space next, Ctrl+Space previous.
V	View Cycle: step through the group's images that are in the current view.
Shift+C	Labeled Cycle: step through the group's images that have labels - for checking or correcting them.
W	Waterfall: the group's images stacked vertically, as a sonar record. Hold Space to glide up, Ctrl+Space down; + and - change the speed.
P	Pan: drag to pan, wheel to zoom; nothing is placed.
R	Ruler: left-drag measures a ground distance. Shift+drag measures from any mode.

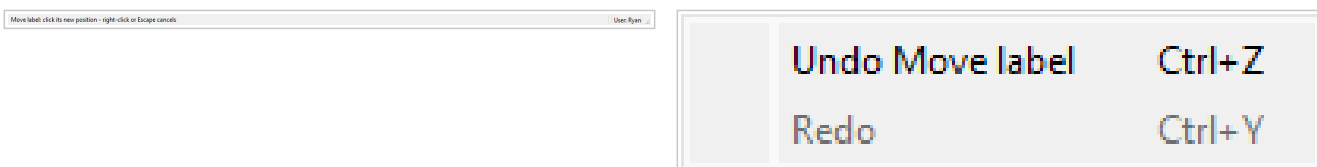


Waterfall mode: the group's images stacked, labels on all of them visible.

Leaving a cycle keeps your place: come back to the mode and it resumes where it was. View > Go to Coordinates... (Ctrl+G) flies to a latitude and longitude; right-click the map in Pan mode to add a Waypoint there, a named place to return to from the Waypoints panel. Waypoints export to GPX for other tools.

7 Correcting labels, and undoing

1. A label in slightly the wrong place: right-click it > Move, then click where it belongs. On its own image only the position changes - masks, heading, size, rating and review stay. A click on another image asks first, because the label would change image and lose what was recorded on this one's pixels.
2. The wrong class: right-click > Change Class, which changes the whole linked object. Or change it in the Snippet Editor, where only this label can be changed if you prefer.
3. Anything else: Edit > Undo (Ctrl+Z) takes back the last edit, Redo (Ctrl+Y) puts it back. The menu names the step. The Snippet Editor's Ctrl+Z works on the same history.

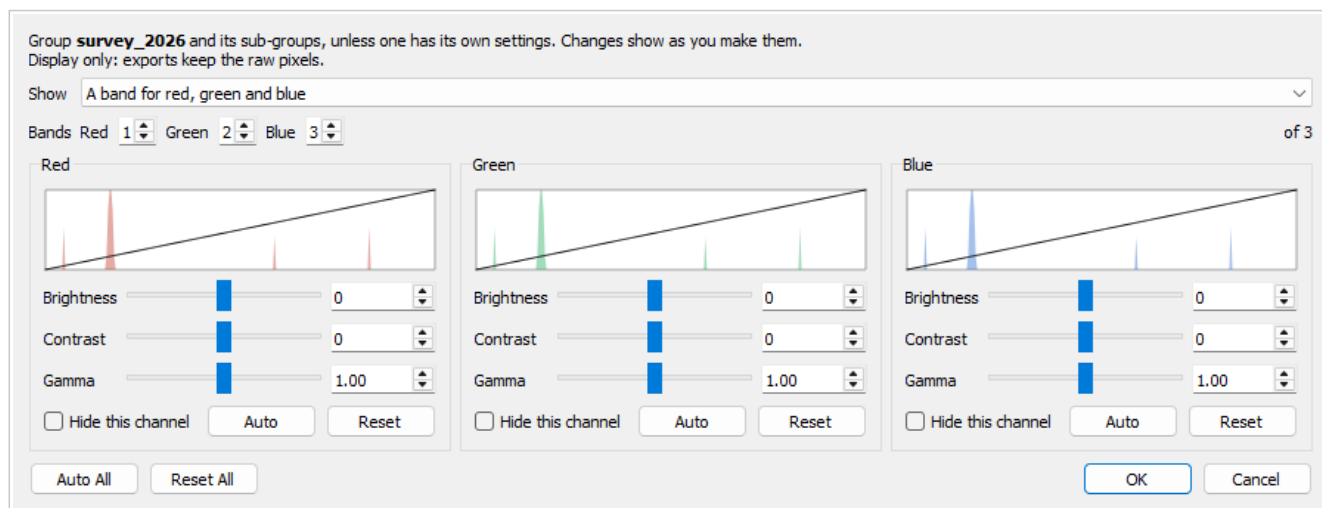


The status bar while a label is being moved, and the Edit menu naming the step Undo would take.

Nearly everything is undoable: placing, removing, linking, moving and describing labels, masks (a step per stroke), headings, sizes, ratings, reviews, notes, class edits, Clear All Labels, Import Ground Truth and Remove Missing Images. Adding imagery and display settings are not. Up to 200 steps are kept; opening another project clears them.

8 How the imagery looks

Multi-band imagery is drawn from the bands you choose, with brightness, contrast and gamma per channel, for the group selected in the Layers panel: View > Display Settings... (Ctrl+D). A single band can be shown as greyscale. The settings are saved with the project and affect only what you see - exports and measurements read the imagery as it is.



View > Display Settings... for a group.

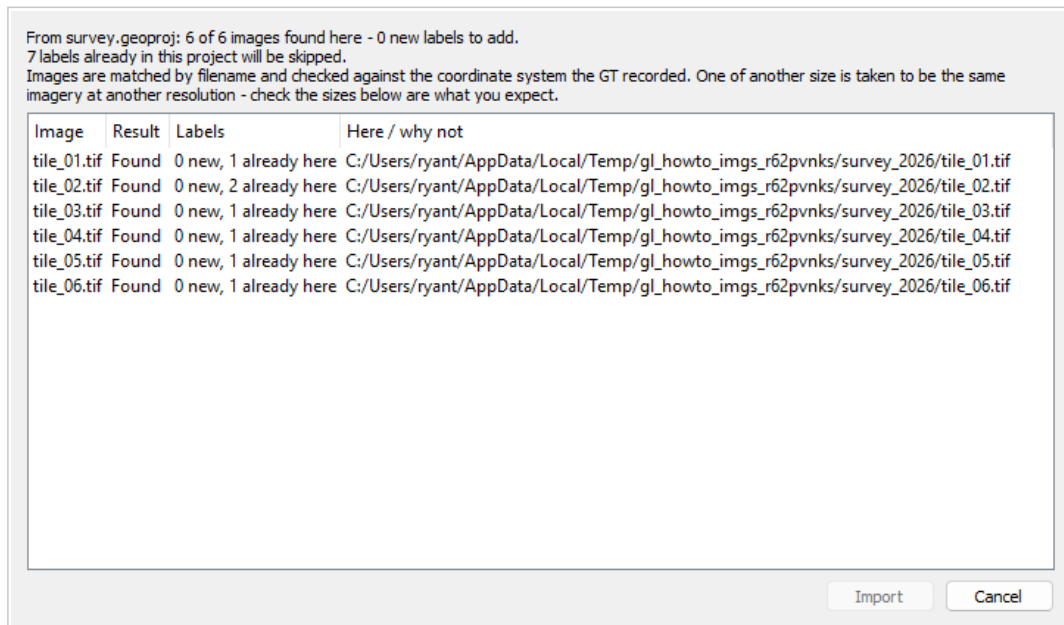
9 Saving, sharing and imagery that has moved

- File > Save Project (Ctrl+S) writes the project file; Save Project As... chooses the name. The title bar shows a * while there is unsaved work. A recovery copy is written as you work, and offered back after a crash.
- A project refers to imagery by path. Opened on another machine, or after the imagery was moved, File > Locate Missing Images... points it at the new folder; File > Remove Missing Images... takes out the images that are not coming back, with their labels (undoable).
- File > Combine Projects... merges another project into this one.
- File > Switch User... sets the name recorded on everything you do, so a shared machine still says who did what.

10 Ground truth in and out

Importing labels made elsewhere

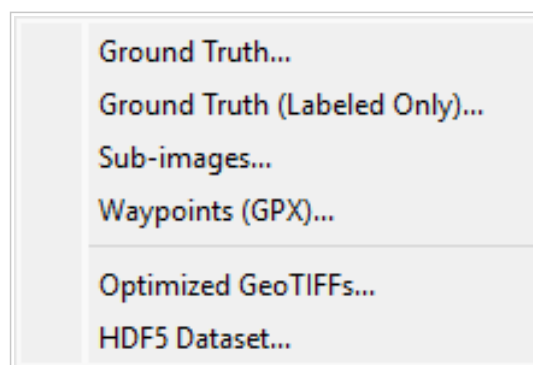
File > Import Ground Truth... adds the labels from another project or ground-truth file onto the images loaded here, matched by filename. The dialog previews what will happen before anything changes: which images were found, which are missing, and how many labels each brings.



The Import Ground Truth preview.

Ground truth made on downsampled copies of the imagery can be imported onto the full-resolution originals: an image of the same name and another size is taken as the same imagery at another resolution, and each label is placed at the same ground position. Masks are left out where sizes differ. Then Labeled Cycle (Shift+C) and Move are the way to check and correct any that are off.

Exporting



The Export menu.

- Export > Ground Truth... writes every label (positions, classes, links, headings, sizes, masks, ratings, reviews) to a file another GeoLabeller - or your own tools - can read. Ground Truth (Labeled Only) leaves out images with no labels.
- Export > Sub-images... cuts a GeoTIFF round every label at a ground size you give.
- Export > HDF5 Dataset... writes the training dataset: a chip per label at a chosen size, with its class, masks, heading, size, rating and provenance, plus negatives. Rejected labels are left out. The dialog estimates what the settings will produce.

- Export > Optimized GeoTIFFs... writes tiled, pyramided copies of the imagery that load far faster next time.
- Export > Waypoints (GPX)... for GPS units and GIS tools.

Extract H x W pixel snippets from the images with a sliding window and write the HDF5 CNN dataset. Snippets containing a label become genuine examples; all others are hard negatives. Existing files are appended to.

Examples (gt=true) from

☐ Layers toggled on, with labels (1)
☒ All loaded layers, with labels (6)
☐ Nothing - hard negatives only

Hard negatives (gt=false) from

☐ Nothing - true positives only
☐ Only the images that contribute examples
☐ Layers toggled on (1)
☐ Flagged hard-negative sources only (0)
☒ All loaded layers (6)

Labelled ground is never written as a hard negative, whichever images are slid.

Snippet options

Snippet height (px): 64
 Snippet width (px): 64
 Overlap: 50 %
 Channels: RGB (3 channels)
 Pixel format: uint8 (stretched 0-255)
☒ Add 8 offset copies of each example
 Example offset: 16 px
 Split (this batch): Train (0)
 Chunk (samples): 1
 Image compression: None

Hard negatives

☐ Split hard negatives across train / validate / test
 Train: 70 % Validate: 15 % Test: 15 %

Output .h5: dataset.h5 (existing file is appended) Browse...

Export Cancel

Export > HDF5 Dataset...

11 Keyboard shortcuts

Help > Keyboard Shortcuts (F1) shows every key. The tables below are taken from the same source, so they match what F1 shows.

Keyboard Shortcuts

File Operations

Ctrl+N New Project
Ctrl+Shift+P Open Project
Ctrl+S Save Project
Ctrl+Shift+S Save Project As
Ctrl+O Add Image (GeoTIFF + custom)
Ctrl+Shift+O Add Directory
Ctrl+Q Exit

Edit

Ctrl+Z Undo the last edit (labels, flags, presets, waypoints)
Ctrl+Y Redo
Ctrl+Shift+Z Redo

Navigation

Mouse Wheel Zoom in/out
Click + Drag Pan (in Pan mode)
Right-click Context menu
Ctrl+G Go to a latitude/longitude
Ctrl+Shift+W Add waypoint by coordinates
Ctrl+D Display Settings for the selected group
Ctrl+B Show/hide the snippet sidebar

Mode Switching

P Pan mode
L Label mode
C Cycle mode (group-based)
V View Cycle (the selected group's images that have labels)
Shift+C Labeled Cycle (the selected group's images that have labels)
W Waterfall mode (stack group vertically)
R Ruler mode

Labeling (Label & Cycle modes)

Left-click Place label
Right-click label Label options (remove, link, describe)
Right-click + drag Pan the view
Ctrl+Left-click label Label options (Cycle modes)
1-9 Quick-switch to class 1-9
Shift+1-9 Description preset 1-9; again for none
Shift+0 No description (new labels get none)
Right-click label > Move Then click its new position
Escape Cancel link mode, or a move

Tips

- Layers default to hidden when loading - expand groups and check to display
- Turning on a layer automatically checks its parent groups
- Add Directory creates a root group named after the selected folder
- Visibility syncs between Layer Panel and Labeled Images Panel
- Custom file readers are auto-detected - registered formats appear in file dialogs

Chain Linking

K Toggle chain-link (any labeling mode)
Left-click label Add to chain (first click anchors)
N Finish this chain, start a new one
Escape Exit chain-link (links are kept)

Cycle Modes (Cycle / View Cycle / Labeled Cycle)

Space Next image (hides the one the cycle showed)
Ctrl+Space Go back to previous layer
Left-click Place label
Right-click label Label options (remove, link, describe)
Right-click + drag Pan around
Mouse wheel Zoom in/out

Waterfall Mode

Hold Space Glide up through the image stack
Hold Ctrl+Space Glide down (starts at the bottom)
+ / - Glide speed up / down
Left-click Place label (group labels stay visible)
Right-click label Label options (remove, link, describe)
Right-click + drag Pan within the stack
Mouse wheel Zoom in/out

Measuring

Shift + Drag Measure distance, without changing mode
Labels > Snippet Editor A label's length and width (Size)
Escape Clear the measurement or the go-to marker

Waypoints

Right-click map Add waypoint here (Pan mode, geo)
Ctrl+Shift+W Add waypoint by coordinate
Right-click marker Go to / Rename / Remove
Double-click in panel Fly to that waypoint
Show on map Hide or show every waypoint marker
Export > Waypoints (GPX) Write every waypoint to a GPX file
File > Import Waypoints... Add the waypoints in a GPX file or another project

Layer Panel

Checkbox Toggle layer/group visibility
Right-click group Select/Unselect, Expand/Collapse all
Right-click layer Zoom to layer, Remove (deletes from project)
Drag & Drop Reorder layers/groups

Labeled Images Panel

Checkbox Toggle image visibility (synced)
Right-click label Zoom to label or layer
Right-click group Select/Unselect all in group

Snippet Editor

Space Next snippet (Single view)
Ctrl+Space Previous snippet (Single view)
1-5 State confidence, 1 lowest; then next, if Advance is ticked
0 Clear the confidence rating
A Review: accept, then next (Single view)
X Review: reject, then next
F Review: look at again, then next
Ctrl+1-9 Label class by its place in the list (whole object)
O / P / G / D / H / V Tool: Orient / Paint masks / Polygon / Wand / Measure / Move label
Click (Wand) Take the connected pixels that look like this one
Right-click (Wand) Take them out of the mask
Drag (Wand) Right for more, left for less, before letting go
Click (Polygon) Add a corner; double-click or Enter fills it
Right-click (Polygon) First click: cut the shape out. After: take back a corner
Backspace / Escape Polygon: take back a corner / give it up
Left-drag Paint a mask, draw a heading, or measure
[/] Brush smaller / larger
H (hold) Look under the masks
Click (Move label) Put the label there, on its own image
Right-drag Erase the mask (Paint)
Right-click Clear the heading (Orient) or the size (Measure)
L / W Measure: the next line is the length / the width
Ctrl+Enter Notes: store the description being typed
Enter Notes: store the Group ID being typed
Escape Notes: put back what was stored
Shift+drag Pan the zoomed snippet
House wheel Zoom to the cursor; kept until Fit
Double-click Open a grid cell in the Single view
Click (Grid, Review) Reject the snippet; again takes it back
Right-click (Grid, Review) Flag the snippet to look at again
A (Grid, Review) Accept the rest of the page, then the next
Ctrl+Z Undo - one history with the main window
Ctrl+Y Redo
Ctrl+S Save Project

Help

F1 Show this help

Close

Help > Keyboard Shortcuts (F1).

File Operations

Key	Does
Ctrl+N	New Project
Ctrl+Shift+P	Open Project
Ctrl+S	Save Project
Ctrl+Shift+S	Save Project As
Ctrl+O	Add Image (GeoTIFF + custom)
Ctrl+Shift+O	Add Directory
Ctrl+Q	Exit

Edit

Key	Does
Ctrl+Z	Undo the last edit (labels, flags, presets, waypoints)
Ctrl+Y	Redo
Ctrl+Shift+Z	Redo

Navigation

Key	Does
Mouse Wheel	Zoom in/out
Click + Drag	Pan (in Pan mode)
Right-click	Context menu

Ctrl+G	Go to a latitude/longitude
Ctrl+Shift+W	Add waypoint by coordinates
Ctrl+D	Display Settings for the selected group
Ctrl+B	Show/hide the snippet sidebar

Mode Switching

Key	Does
P	Pan mode
L	Label mode
C	Cycle mode (group-based)
V	View Cycle (the selected group's images on screen)
Shift+C	Labeled Cycle (the selected group's images that have labels)
W	Waterfall mode (stack group vertically)
R	Ruler mode

Labeling (Label & Cycle modes)

Key	Does
Left-click	Place label
Right-click label	Label options (remove, link, describe)
Right-click + drag	Pan the view
Ctrl+Left-click label	Label options (Cycle modes)
1-9	Quick-switch to class 1-9
Shift+1-9	Description preset 1-9; again for none
Shift+0	No description (new labels get none)
Right-click label > Move	Then click its new position
Escape	Cancel link mode, or a move

Chain Linking

Key	Does
K	Toggle chain-link (any labeling mode)
Left-click label	Add to chain (first click anchors)
N	Finish this chain, start a new one
Escape	Exit chain-link (links are kept)

Cycle Modes (Cycle / View Cycle / Labeled Cycle)

Key	Does
Space	Next image (hides the one the cycle showed)
Ctrl+Space	Go back to previous layer
Left-click	Place label
Right-click label	Label options (remove, link, describe)
Right-click + drag	Pan around
Mouse wheel	Zoom in/out

Waterfall Mode

Key	Does
Hold Space	Glide up through the image stack
Hold Ctrl+Space	Glide down (starts at the bottom)
+ / -	Glide speed up / down
Left-click	Place label (group labels stay visible)
Right-click label	Label options (remove, link, describe)
Right-click + drag	Pan within the stack
Mouse wheel	Zoom in/out

Measuring

Key	Does
Shift + Drag	Measure distance, without changing mode
Labels > Snippet Editor	A label's length and width (Size)
Escape	Clear the measurement or the go-to marker

Waypoints

Key	Does
Right-click map	Add waypoint here (Pan mode, geo)
Ctrl+Shift+W	Add waypoint by coordinate
Right-click marker	Go to / Rename / Remove
Double-click in panel	Fly to that waypoint
Show on map	Hide or show every waypoint marker
Export > Waypoints (GPX)...	Write every waypoint to a GPX file
File > Import Waypoints...	Add the waypoints in a GPX file or another project

Layer Panel

Key	Does
Checkbox	Toggle layer/group visibility
Right-click group	Select/Unselect, Expand/Collapse all
Right-click layer	Zoom to layer, Remove (deletes from project)
Drag & Drop	Reorder layers/groups

Labeled Images Panel

Key	Does
Checkbox	Toggle image visibility (synced)
Right-click label	Zoom to label or layer
Right-click group	Select/Unselect all in group

Help

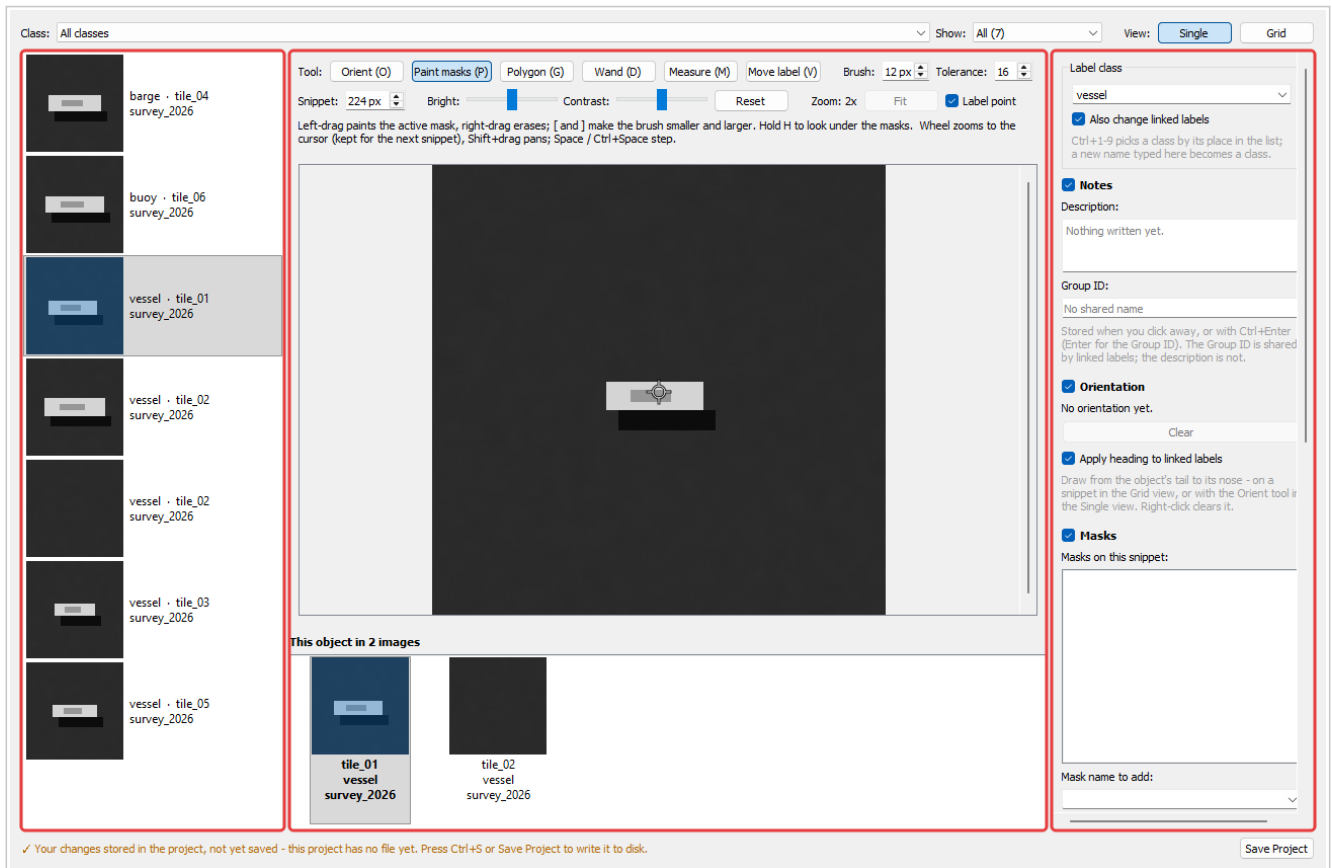
Key	Does
F1	Show this help

Part 2

The Snippet Editor

12 What it is for

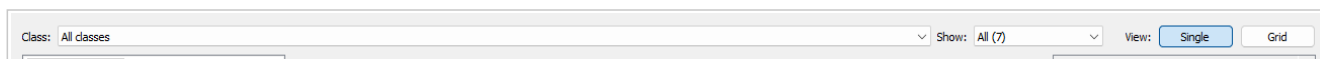
A label on the map is a point and a class. The Snippet Editor is where everything else about the object is recorded, one snippet at a time: its outline as masks, its heading, its length and width, how sure you are of it, whether it has been reviewed, and any notes. Open it with Labels > Snippet Editor.... It shows the labels of the open project; what you do in it is stored in the project as you go, and the bar at the bottom says whether that is on disk yet.



The Snippet Editor. Left: the list of snippets. Middle: the Single view - one snippet, zoomable, with the tools. Right: the column of sections, one per thing recorded.

Each section on the right switches on and off by its header. Switching one off takes all of it away - its panel, its entry on the Show filter, its overlay on the snippets, its tool and its keys - so the editor shows only what your project records.

13 The list, and stepping through it

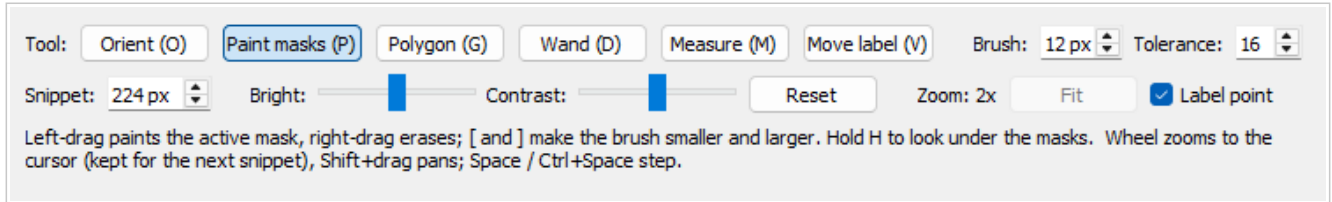


Class chooses which labels are listed; Show narrows them to what still needs doing; View switches Single and Grid.

- CLASS lists one class, all classes, or one linked object (every object is offered after the classes, named by its class, group ID and number of views).
- SHOW offers, for every section that is on, the snippets that still need it and the ones that have it: Needs masks / Masked, Needs orientation / Oriented, and so on, with live counts. A finished snippet leaves a Needs list when you move off it, so the one in hand never vanishes under you.
- SPACE steps to the next snippet and Ctrl+Space to the previous, wherever the keyboard focus is. Clicking a row works too.
- Thumbnails are read as they come into view, so a project of thousands of labels opens at once.

14 Masks

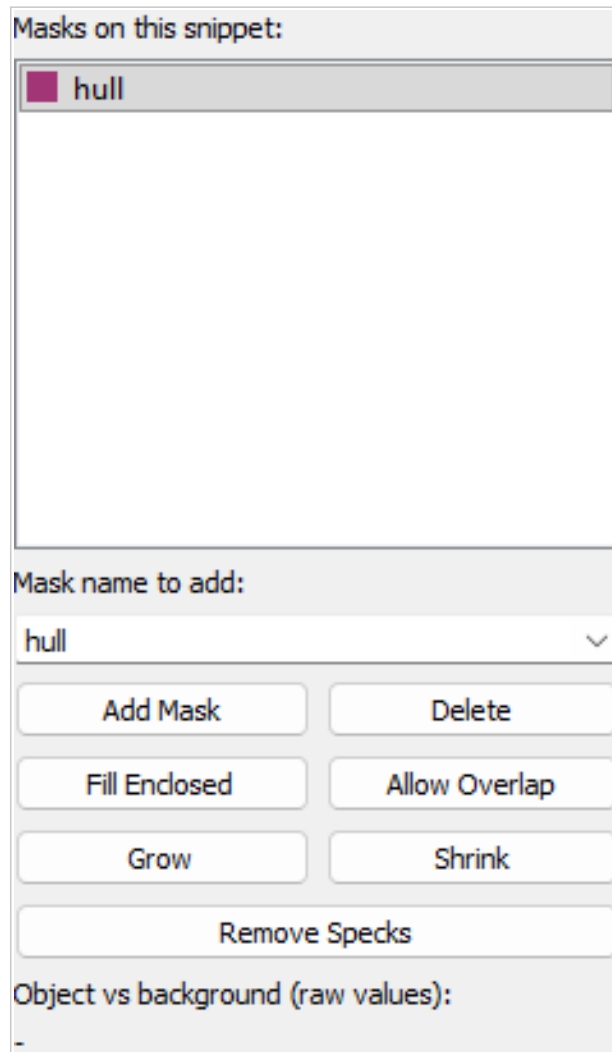
A mask is a named binary layer painted over the snippet - 'hull', 'shadow' - stored in the project anchored to the imagery, so it stays put whatever snippet size you look at it in. Several masks can sit on one snippet; by default they do not overlap, so other masks act as walls for whatever you draw.



The tool row (one button per tool, its key in brackets), the view row, and the line that says what the tool in hand does.

Adding a mask

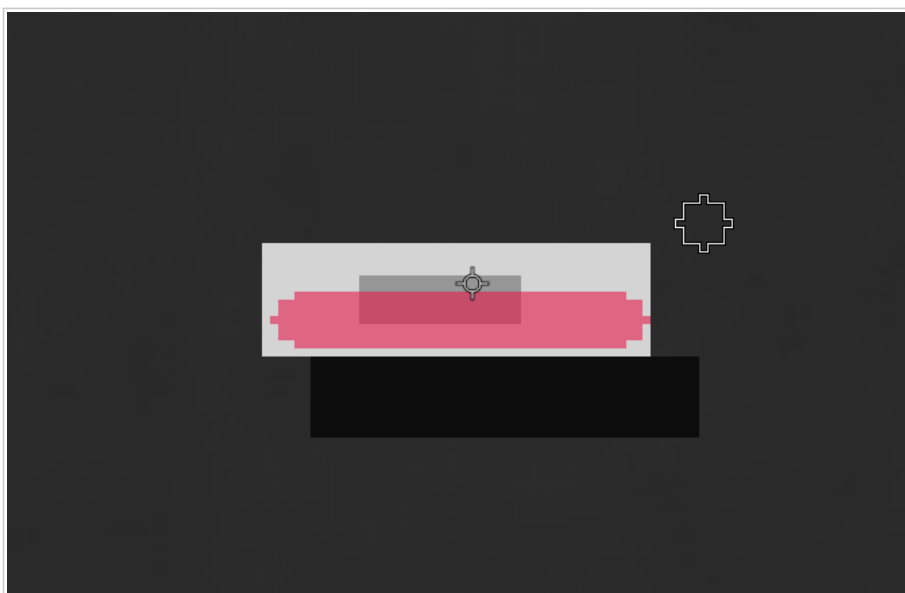
1. In the Masks section, pick or type the name under 'Mask name to add' and press Add Mask. The name joins the project's presets (Labels > Edit Mask Names...), so the next snippet offers it.
2. The new mask is the active one - the one the tools paint. Choosing a name the snippet already has selects that mask.



The Masks section.

Paint masks (P)

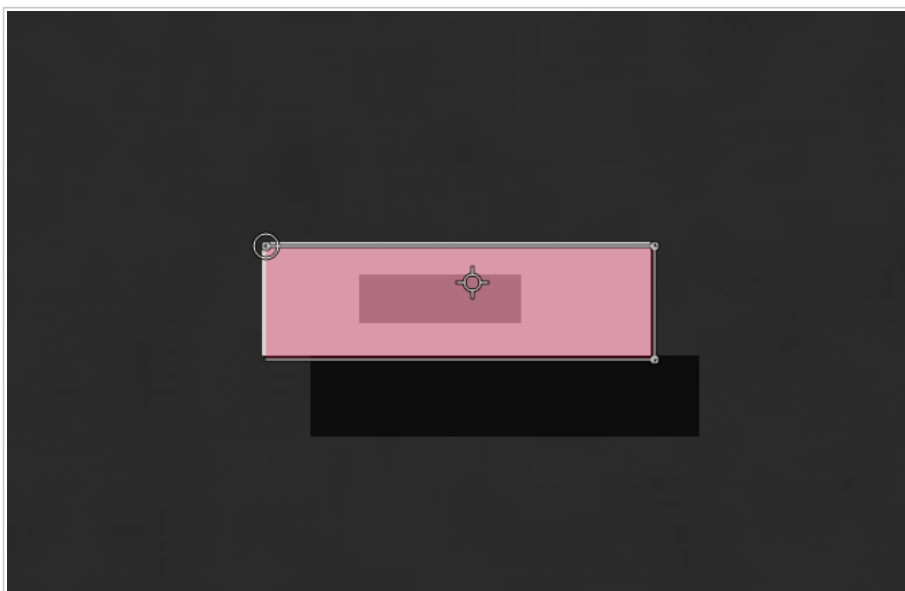
Left-drag paints the active mask, right-drag erases. The outline under the cursor shows exactly which pixels the next stamp will paint. [and] make the brush smaller and larger.



A brush stroke, and the brush outline.

Polygon (G)

1. Click round the object, corner by corner. The shape fills in as you go.
2. Double-click, press Enter, or click the first corner to finish: the shape is filled into the active mask.
3. Right-click or Backspace takes back the last corner; Escape gives the shape up.
4. To cut a shape OUT of the mask, begin it with a right-click.

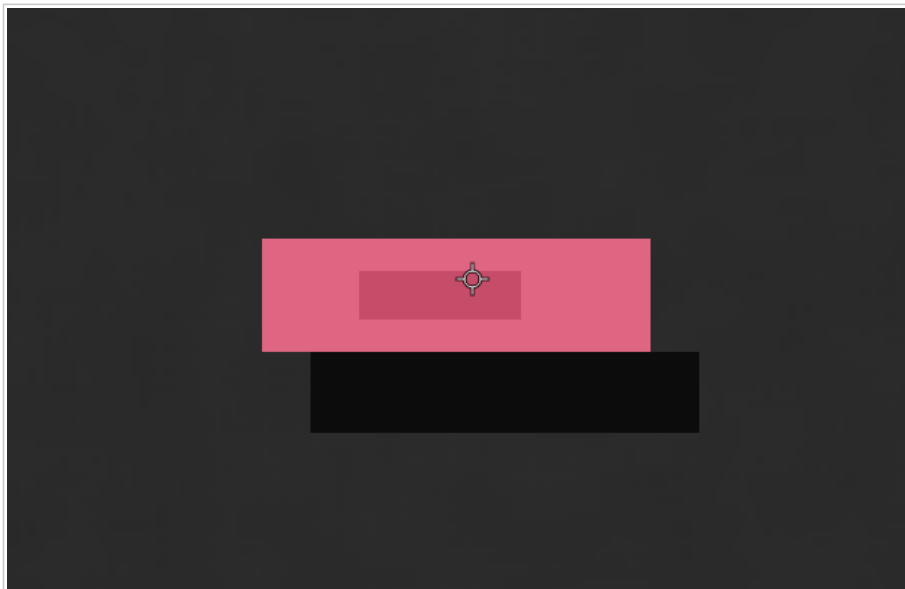


A polygon being clicked out: three corners placed, the fourth following the cursor, the ring marking where a click closes it.

Wand (D)

1. Click the object. The connected pixels that look like the one you clicked light up - then join the active mask when you let go. A right-click takes them out of the mask instead.

2. Too little or too much? Keep the button down and drag right to take in more, left to take in less. The Tolerance box follows, and the next click starts from where you ended.

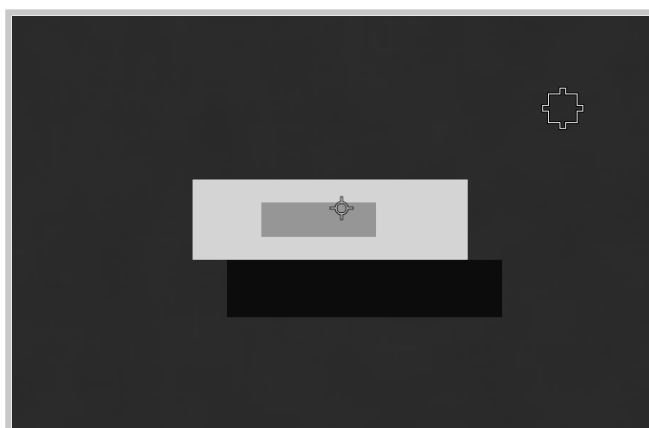
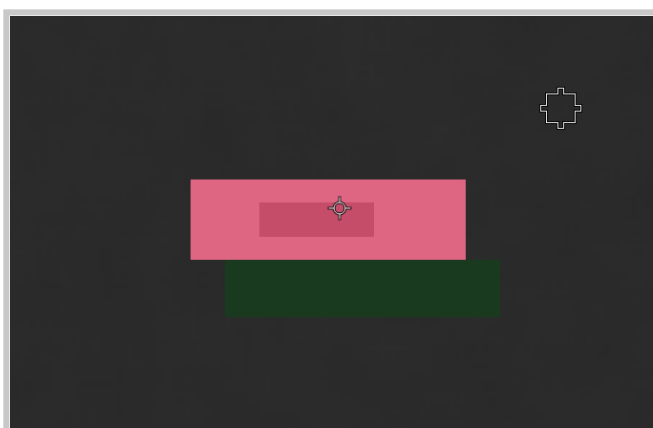


The wand held down and dragged right: the hull and its wheelhouse are both within tolerance now and shown as what will be taken.

'Looks like' is judged on the snippet as you see it, brightness and contrast included. Raise the contrast and the wand tells apart what it otherwise would not.

Tidying a mask

- GROW and SHRINK make the active mask one pixel wider or narrower all round.
- REMOVE SPECKS drops the separate pieces smaller than the brush - set the brush size to say how small. The largest piece always stays, and the button says what it found.
- FILL ENCLOSED fills the holes: the inside of a drawn outline, or what the wand left out. It refuses when the outline has a gap.
- Grow, Fill Enclosed, Shrink closes a ragged mask in three clicks.
- ALLOW OVERLAP lets masks share pixels; off, other masks are walls for everything you draw, and edges for Fill Enclosed.
- Hold H to look at the imagery under the masks.



The same snippet with its two masks drawn, and with H held.

Masks on this snippet:

	hull
	shadow

Mask name to add:

hull 

Add Mask Delete

Fill Enclosed Allow Overlap

Grow Shrink

Remove Specks

Object vs background (raw values):

'hull': 672 px object, 49504 px background
B1: obj 200.9±23.7 bg 42.2±3.0
B2: obj 200.9±23.7 bg 42.2±3.0
B3: obj 200.9±23.7 bg 42.2±3.0

Two masks on a snippet - hull and shadow - and the tidying buttons. Below them, the object's raw pixel values against the background.

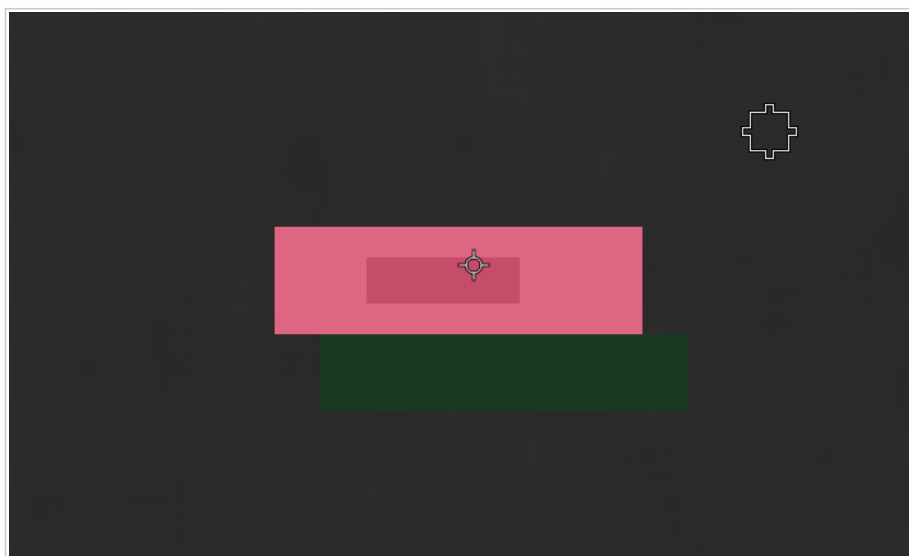
Every stroke, shape, wand click and tidy is one undo step, and is stored in the project the moment it is made.

15 Looking closely



The view row: the snippet size, brightness and contrast (display only), the zoom with Fit, and the Label point.

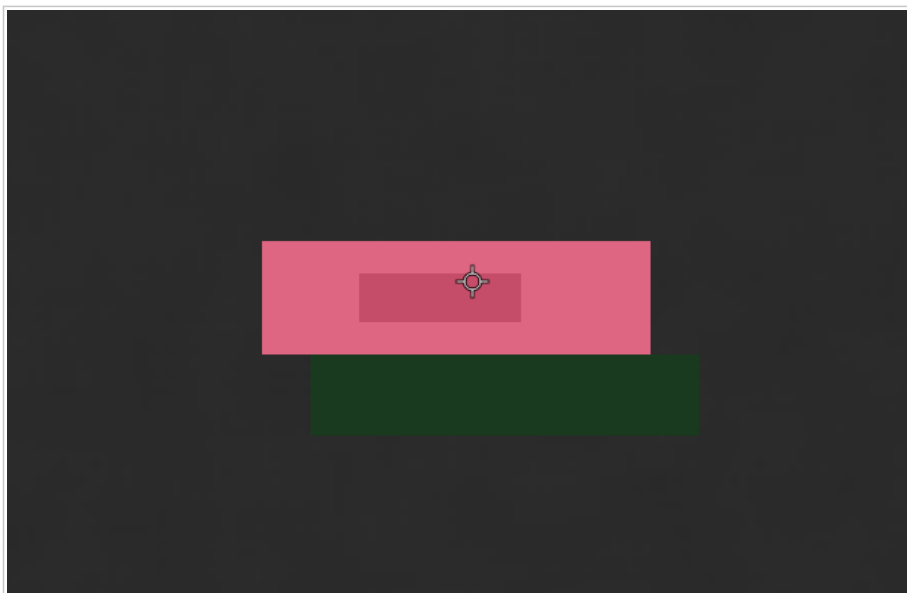
- The WHEEL zooms to the cursor, 0.25x to 32x; Shift+drag pans. The zoom you set stays for the next snippet, with the view held on the label, so a pass of fine work needs no re-zooming. FIT goes back to the whole snippet.
- SNIPPET is how much of the image is shown round the label, in source pixels. Masks are anchored to the imagery, so changing it only changes the window you work in.
- BRIGHT and CONTRAST change the display only; Reset puts them back.
- LABEL POINT marks where the label itself is - a ring, open in the middle. Near an image's edge that is not the middle of the snippet.



Zoomed in on the hull.

Move label (V)

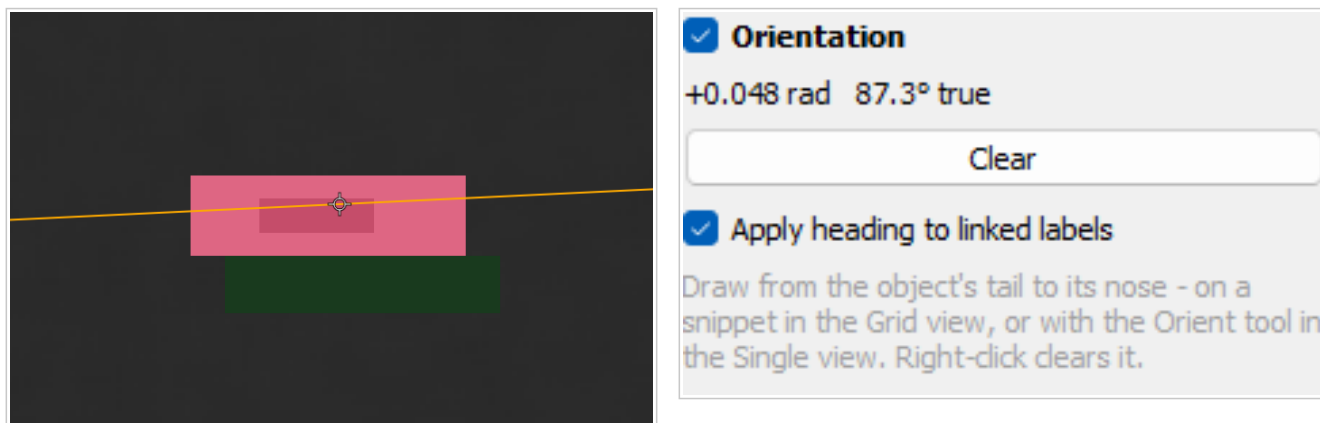
Click where the label belongs and it moves there, on its own image, to a fraction of a pixel when zoomed. Masks, heading, size, rating and review stay as they were. The tool stays in hand, so correcting a run of positions is Space, click, Space, click.



The Move label tool: the ring is the label's own point; a click puts it elsewhere.

16 Orientation

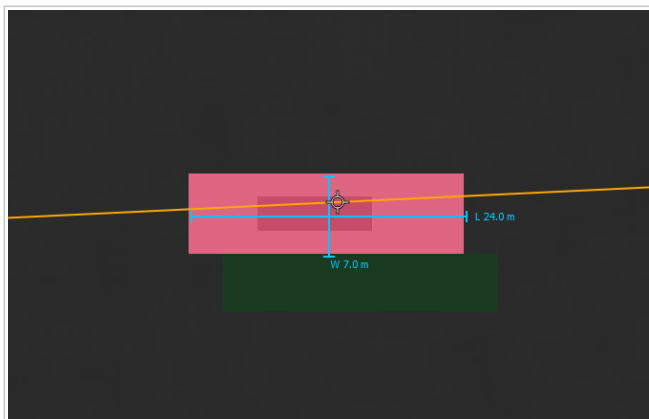
With the Orient tool (O), drag across the snippet from the object's tail to its nose. The pixel angle is stored, and for georeferenced imagery the true-north heading too. Right-click clears it. The Orientation section reads out the heading, and with 'Apply heading to linked labels' ticked the other views of a linked object take the same true-north heading, each worked out through its own image (shown violet until drawn over by hand).



An orientation drawn with the Orient tool, and the Orientation section.

17 Size

With the Measure tool (M), draw the length across the object, then the width: the first line is the length, the next the width, in metres on the ground. L and W choose which the next line sets; right-click clears both. The Size section shows them and can share them with the linked labels.



☒ **Size**

With the Measure tool, draw the length, then the width. The next line sets:

☐ Length: 24.0 m

☒ Width: 7.0 m

☐ Apply size to linked labels

Clear

Metres on the ground, measured on the ellipsoid.
Right-click clears; L and W choose the line.

A length and a width measured, and the Size section.

18 Confidence

How sure you are of the label, 1 (lowest) to 5 (highest): the buttons, or keys 1 to 5. 0 clears it. Unrated labels export as 0. Tick 'Advance after rating' and a rating key moves on to the next snippet, so a rating pass is one key a snippet.

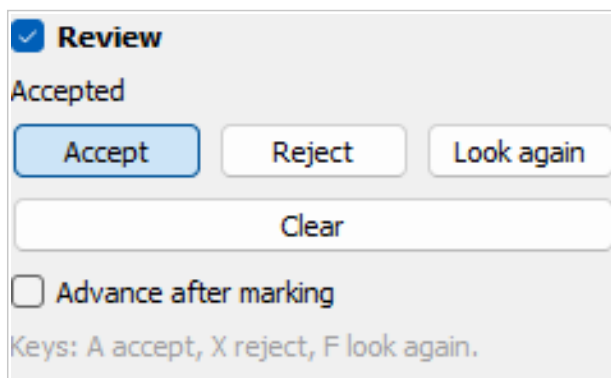


The screenshot shows a user interface for the 'Confidence' section. At the top, there is a checked checkbox labeled 'Confidence'. Below it, the text reads 'How sure are you of this label? 1 lowest, 5 highest.' Underneath this text are five buttons labeled '1', '2', '3', '4', and '5'. The button labeled '4' is highlighted with a blue border. Below the buttons is a 'Clear' button. At the bottom, there is an unchecked checkbox labeled 'Advance after rating'. Below this checkbox, the text reads 'Keys 1-5 rate the snippet in hand; 0 clears. Unrated labels export as 0.'

The Confidence section.

19 Review

Whether somebody has checked the label, and what they found: Accept (A), Reject (X) or Look again (F). With 'Advance after marking' on, a key also moves to the next snippet. A rejected label stays in the project and the ground truth, is drawn dimmed on the map, and is left out of the HDF5 and sub-image exports. For reviewing many at once, see the Grid view's Review mode (section 22).



The screenshot shows a software interface for reviewing labels. At the top, there is a checked checkbox followed by the word "Review". Below this, the word "Accepted" is displayed. Underneath "Accepted" are three buttons: "Accept" (highlighted with a blue border), "Reject", and "Look again". Below these buttons is a "Clear" button. Further down, there is an unchecked checkbox followed by the text "Advance after marking". At the bottom of the panel, a line of text reads: "Keys: A accept, X reject, F look again."

The Review section.

20 Notes

The label's description and its object's Group ID, shown and edited in place. The description is this label's alone; the Group ID is one name for every label linked to it. A note is stored when you click away, or with Ctrl+Enter (Enter for the Group ID); Escape puts back what was stored.

☒ **Notes**

Description:

Listing to port; wake visible.

Group ID:

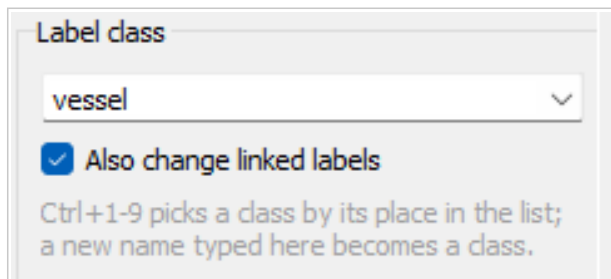
tug 7

Stored when you click away, or with Ctrl+Enter (Enter for the Group ID). The Group ID is shared by linked labels; the description is not.

The Notes section.

21 Changing the class

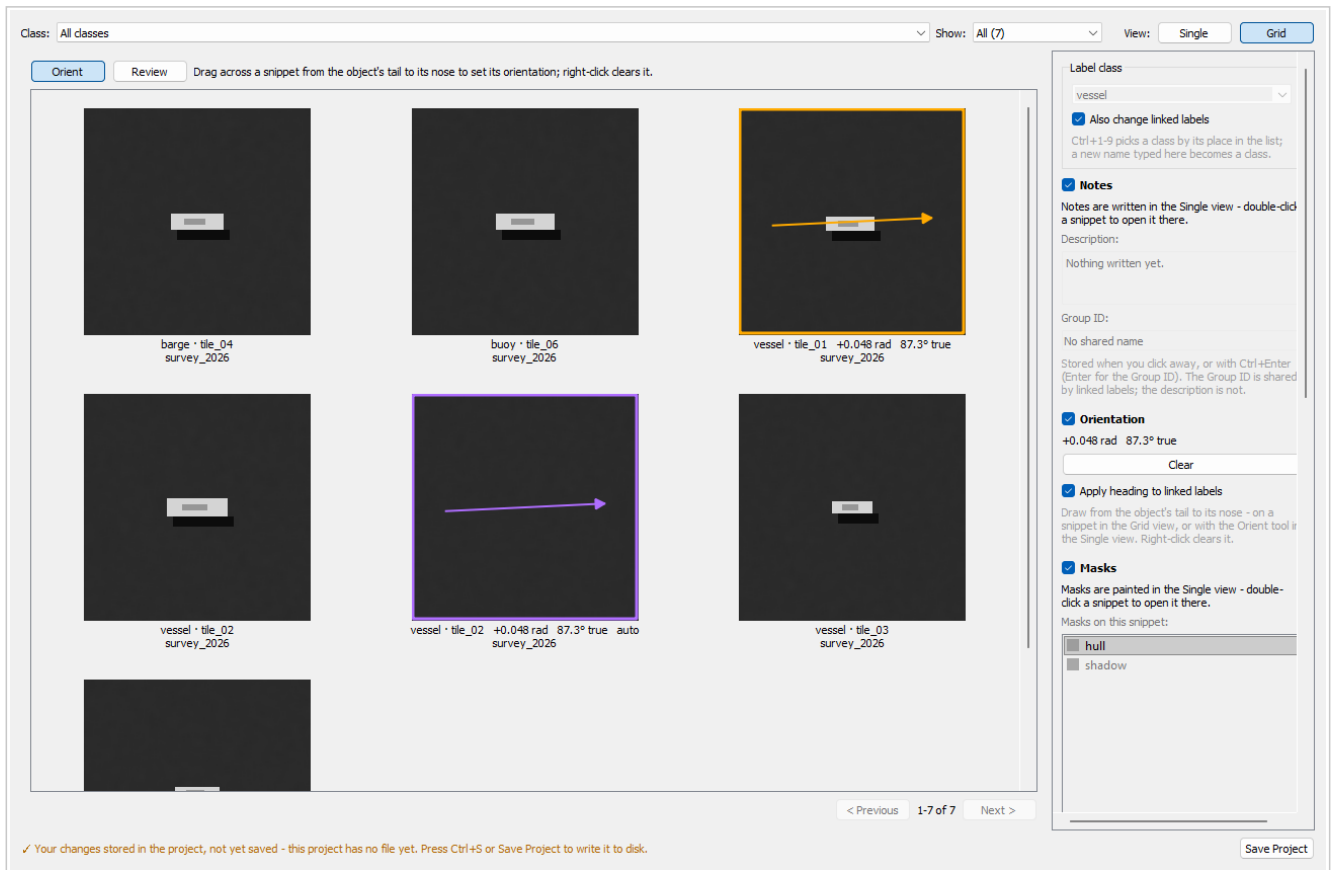
The Label class box gives the snippet in hand another class; Ctrl+1 to Ctrl+9 pick by place in the class list. A new name typed here becomes a class. With 'Also change linked labels' ticked the whole linked object changes; unticked, only this snippet does - for the part of an object that needs a label of its own.



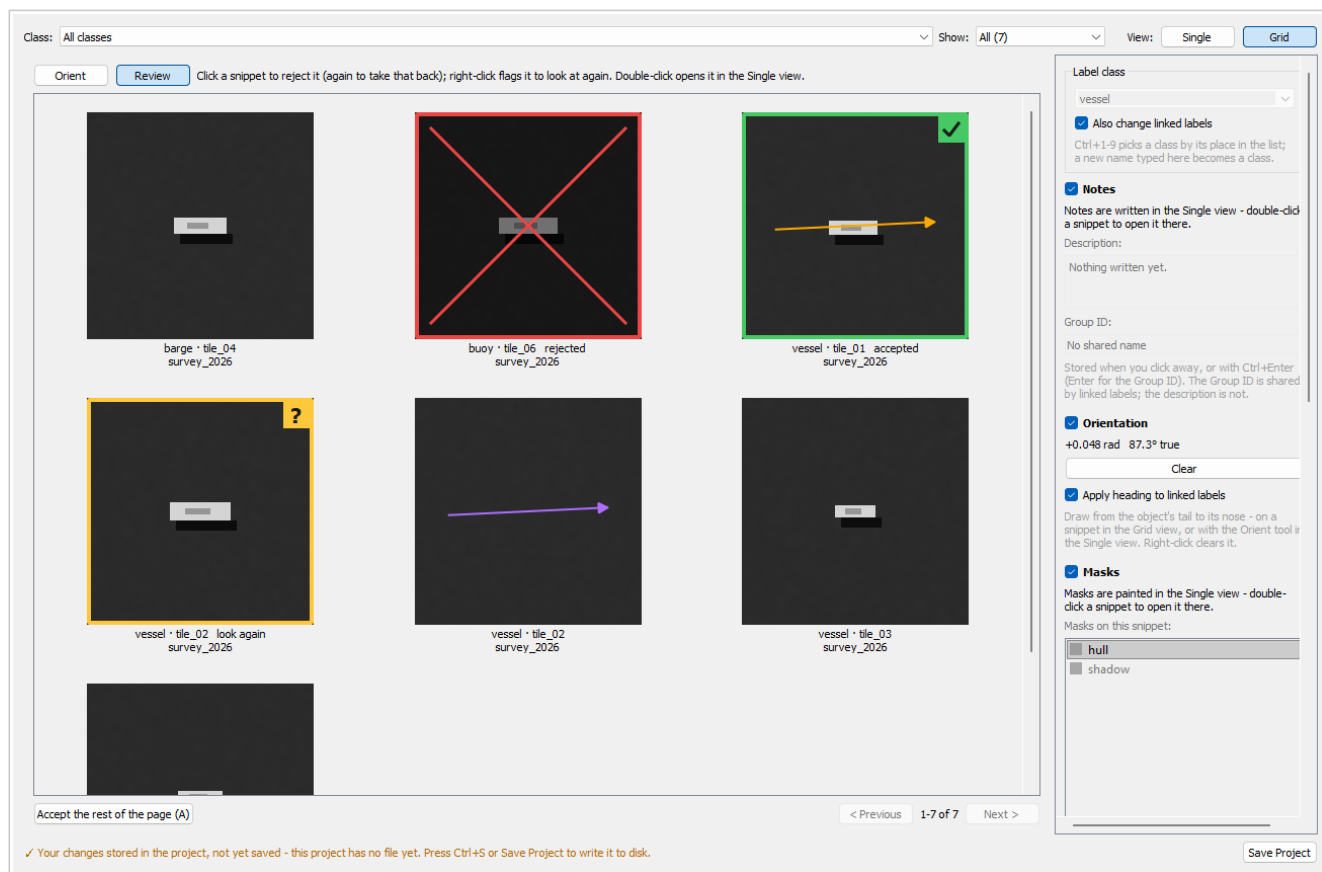
The Label class box.

22 The Grid view

The Grid view lays out a page of sixty snippets at once, in the list's order. It has two modes. In ORIENT mode, drag across a snippet to set its heading - a whole page oriented without opening any. In REVIEW mode each snippet wears what its review found; click one to reject it (again to take that back), right-click to flag it to look at again, and press A - or Accept the rest of the page - to accept everything left unmarked and go on to the next page. Double-click any snippet to open it in the Single view.



The Grid view in Orient mode.



The Grid view in Review mode: one snippet rejected, one flagged, the rest ready for 'Accept the rest of the page'.

Accepting a page is one undo step.

23 One object, every view

When the snippet in hand belongs to a linked object, a row under the Single view shows every view of that object, captioned with its image and what has been recorded on it. Click one to put it in hand. Choosing the object in the Class box lists only its views.



The linked row under the Single view.

24 Saving, and the keys

Everything you do in the editor is stored in the project at once; the bar at the bottom says whether the project is on disk yet. Save Project (Ctrl+S) writes it, from either window.

✓ Your changes stored in the project, not yet saved - this project has no file yet. Press Ctrl+S or Save Project to write it to disk. Save Project

The save bar.

Every key the editor takes:

Key	Does
Space	Next snippet (Single view)
Ctrl+Space	Previous snippet (Single view)
1-5	Rate confidence, 1 lowest; then next, if Advance is ticked
0	Clear the confidence rating
A	Review: accept, then next (Single view)
X	Review: reject, then next
F	Review: look at again, then next
Ctrl+1-9	Label class by its place in the list (whole object)
O / P / G / D / M / V	Tool: Orient / Paint masks / Polygon / Wand / Measure / Move label
Click (Wand)	Take the connected pixels that look like this one
Right-click (Wand)	Take them out of the mask
Drag (Wand)	Right for more, left for less, before letting go
Click (Polygon)	Add a corner; double-click or Enter fills it
Right-click (Polygon)	First click: cut the shape out. After: take back a corner
Backspace / Escape	Polygon: take back a corner / give it up
Left-drag	Paint a mask, draw a heading, or measure
[/]	Brush smaller / larger
H (hold)	Look under the masks
Click (Move label)	Put the label there, on its own image
Right-drag	Erase the mask (Paint)
Right-click	Clear the heading (Orient) or the size (Measure)
L / W	Measure: the next line is the length / the width
Ctrl+Enter	Notes: store the description being typed
Enter	Notes: store the Group ID being typed
Escape	Notes: put back what was stored
Shift+drag	Pan the zoomed snippet
Mouse wheel	Zoom to the cursor; kept until Fit
Double-click	Open a grid cell in the Single view
Click (Grid, Review)	Reject the snippet; again takes it back
Right-click (Grid, Review)	Flag the snippet to look at again
A (Grid, Review)	Accept the rest of the page, then the next
Ctrl+Z	Undo - one history with the main window
Ctrl+Y	Redo
Ctrl+S	Save Project